HIRELOOP

India-first AI recruiting platform

Two AI agents. One candidate graph. Built for the Indian job market.

Build overview · Architecture · Status · Scale

Prepared for team review · June 2026

AT A GLANCE

What we're building

Hireloop is a two-sided, AI-run recruiting platform for India. Candidates are guided end-to-end by Aarya — an AI career partner that ingests their LinkedIn, CV and a short voice call, then surfaces real, scored job matches. Recruiters are served by Nitya, which drafts warm intros and runs their pipeline. A shared Postgres candidate graph connects both sides.

Aarya

candidate AI agent

Nitya

recruiter AI agent

23 build steps mapped

8 steps built

30+ database tables

12 external tools / APIs

THE BIG PICTURE

A two-sided marketplace, run by AI

CANDIDATE SIDE

Aarya

- Signup via LinkedIn
- Auto-built profile (LinkedIn + CV)
- 15-min voice / text career chat
- Scored, explained job matches
- One-tap “Request intro”
- Tailored résumé + mock interview

Shared candidate graph

`Supabase Postgres + pgvector`
`embeddings · RLS · realtime`

RECRUITER SIDE

Nitya

- Recruiter signup + role intake
- Nitya generates the hiring brief
- Per-role candidate search
- Kanban pipeline of candidates
- Warm intros drafted + Gmail-sent
- Inbox + status handshake

THE ENGINE

Two agents on one LangGraph runtime

Aarya

Candidate-facing career partner

TOOLS

- `profile_read`
- `build_career_path`
- `job_search`
- `get_match_score`
- `request_intro`
- `save_job / direct_apply`

Nitya

Recruiter & hiring-manager agent

TOOLS

- `candidate_lookup`
- `draft_email`
- `send_via_gmail`
- `update_intro_status`
- `pipeline ops`
- `role brief generation`

Agent runtime

`LangGraph 1.x`

single-threaded master loop, Postgres checkpoints

`OpenRouter · Claude`

LLM reasoning + tool-calling

`Deepgram → Sarvam`

voice STT + TTS (Sarvam planned, Indian langs)

`agent_actions`

every tool call → table → Realtime UI counter

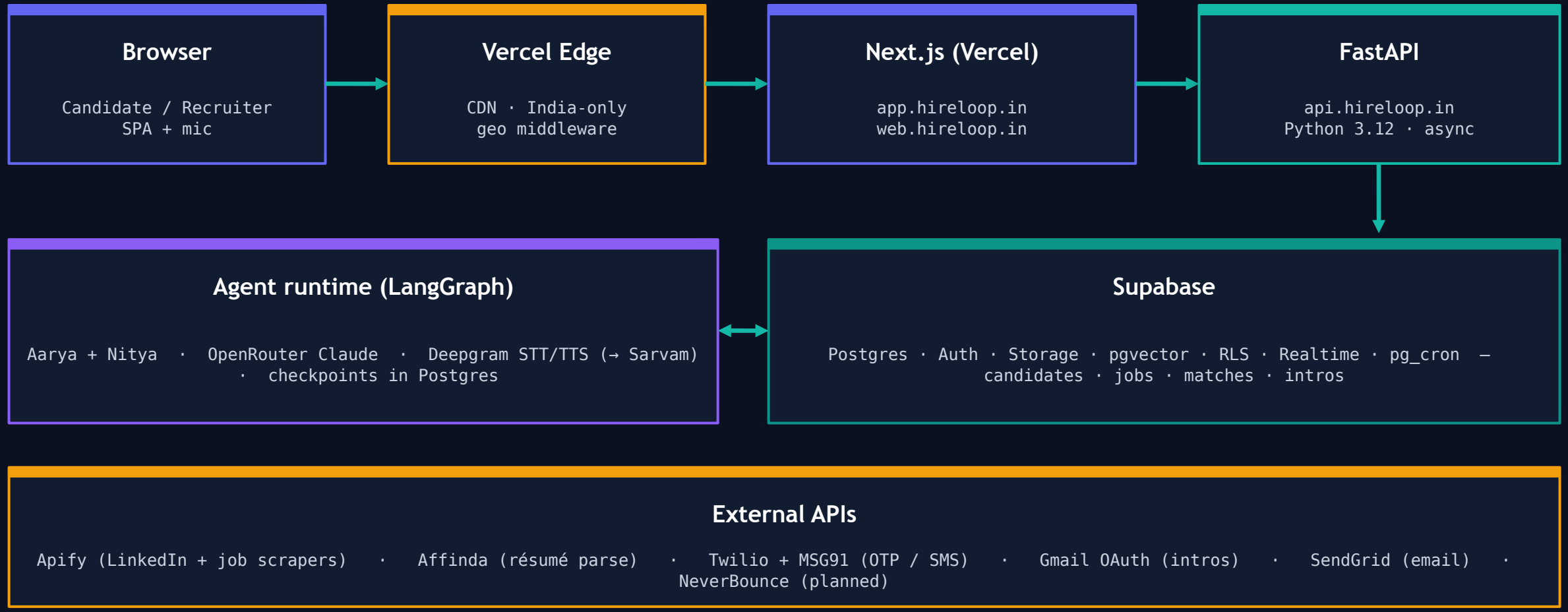
TECHNOLOGY

The full stack, by layer

Frontend	Next.js 15 · React 18 · TypeScript (strict) · Tailwind · Radix UI · Zustand · React-Hook-Form + Zod · Framer Motion
Backend API	FastAPI · Python 3.12 · Pydantic v2 · asyncpg · httpx · structlog · Uvicorn · python-jose
AI / Agents	LangGraph 1.x · langchain-openai · OpenRouter (Claude) · Deepgram STT/TTS now → Sarvam AI planned (Indian langs)
Data	Supabase Postgres · pgvector (HNSW, cosine) · Row-Level Security · Realtime · Storage buckets · pg_cron
External services	Apify (LinkedIn + jobs) · Affinda (résumé) · Twilio + MSG91 (OTP / SMS) · Gmail OAuth (intros) · SendGrid (email) · NeverBounce (planned)
Hosting & infra	Vercel (app + web · edge CDN + India geo) · Supabase (Postgres · Auth · Storage · Realtime) · GitHub Actions CI/CD

ARCHITECTURE

How requests flow, end to end



DATA MODEL

30+ tables across six domains

Identity & consent

users
candidates
recruiters
consent_log
dpdp_export_jobs

Candidate profile

resumes
career_paths
candidate_embeddings
voice_sessions
saved_jobs

Jobs & matching

jobs
companies
job_embeddings
match_scores
match_audits
job_ingest_log

Intros & hiring

intro_requests
hiring_managers
gmail_tokens
job_applications
placements

Comms & interviews

conversations
messages
agent_actions
mock_interviews
notifications
whatsapp_messages
tailored_resumes

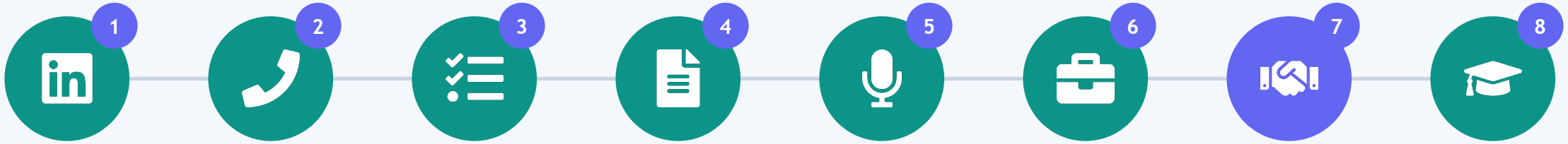
Recruiter workspace

roles
role_versions
role_pipeline
recruiter_searches

*20 SQL migrations · pgvector HNSW (cosine) on every *_embedding · Row-Level Security on every table · soft-delete on PII*

USER JOURNEY — CANDIDATE

From LinkedIn click to a warm intro



1	2	3	4	5	6	7	8
Sign up	**Verify**	**Onboard**	**Profile**	**Voice call**	**Matches**	**Request intro**	**Level up**
LinkedIn OAuth	+91 phone OTP	goals + consent	LinkedIn + CV parse	15-min with Aarya	scored + explained	the core loop	tailored CV + mock

What makes it different

Zero-form onboarding
— profile is built from LinkedIn + CV + voice, not typed.

Honest, explained matches
— every role shows a score and why it fits, India-only.

USER JOURNEY — RECRUITER

From open role to candidate intros

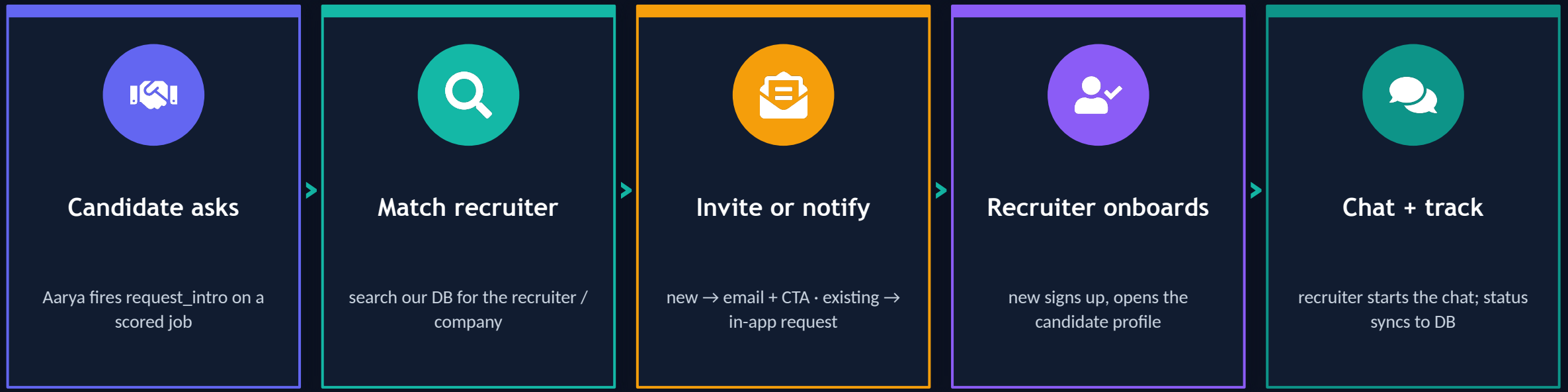
1	2	3	4	5	6	7
Sign up	**Role intake**	**Search**	**Pipeline**	**Draft intro**	**Send**	**Inbox**
recruiter role	Nitya hiring brief	per-role candidates	kanban stages	Nitya writes it	via Gmail OAuth	threads + status

Recruiters are pulled in on demand

When a candidate requests an intro, the platform looks the recruiter up in our database. Already on Hireloop? They get an in-app intro request. New to us? They get a one-click email invite to sign up, see the candidate, and start the chat. The database — never agent-to-agent calls — keeps both sides in sync. No cold email, ever.

THE CORE LOOP

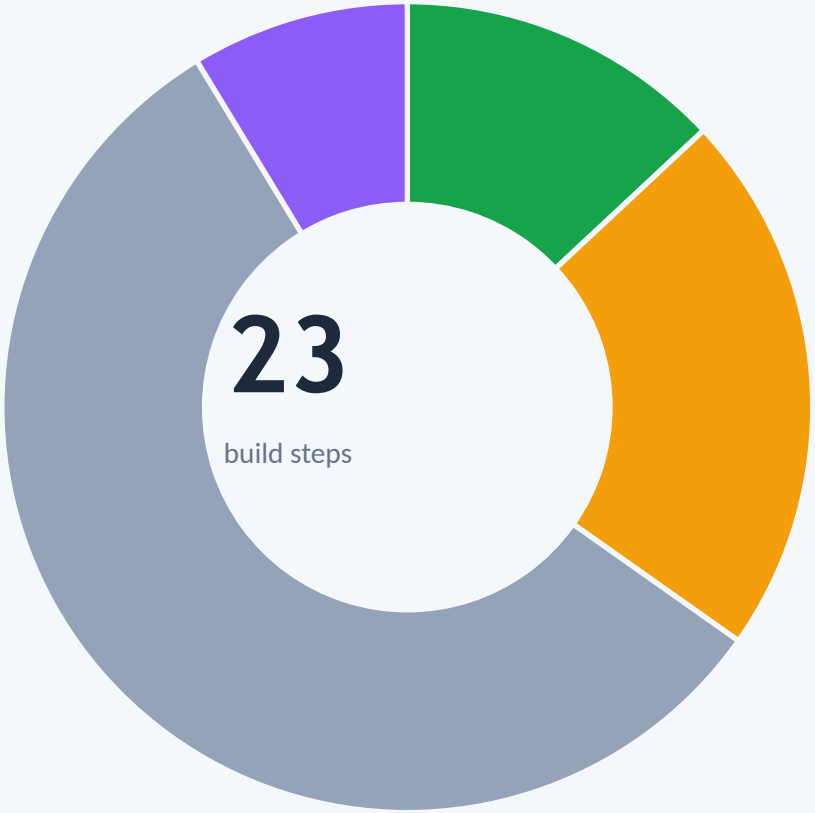
Intro handshake — the MVP-critical path



DB-driven — recruiters are pulled in on demand: a new recruiter gets a one-click email invite, an existing one an in-app request. No cold email, no agent-to-agent RPC.

STATUS

Where the build stands today



3 Done

LinkedIn signup · onboarding wizard · résumé upload + parse

5 Improving

Phone OTP (verify / alt) · Aarya text + voice · matching · match feed

13 Pending

S07 job ingestion (Apify ready) · S10–S21 (intros, recruiter side, admin, SEO)

2 Planned

S22 deploy (Vercel + Supabase) · S23 payments (v2)

STATUS DETAIL

Build status — all 23 steps, in order

Done · Improving · Pending · Planned

S01	LinkedIn signup	S09	Job match feed UI	S17	Pipeline kanban
S02	Phone OTP — verify / alt	S10	Hiring-manager enrichment	S18	Recruiter inbox + Nitya
S03	Onboarding wizard	S11	Gmail intros	S19	Admin panel + DPDP
S04	Aarya text chat	S12	Intros inbox	S20	Transactional email
S05	Aarya voice session	S13	Mock interview	S21	Programmatic SEO site
S06	Résumé upload + parse	S14	Tailored résumé per JD	S22	Deploy — Vercel + Supabase
S07	Job ingestion (Apify)	S15	WhatsApp notify	S23	Payments (v2)
S08	Embeddings + matching	S16	Recruiter signup + roles		

STATUS DETAIL

In progress, pending & deferred

EXTERNAL DEPENDENCIES

Step	Needs	Status
S07 Job ingestion	APIFY_TOKEN	In hand — wiring & running now
S02 Phone OTP	Twilio / MSG91	Implemented but flaky — verify / replace
S11 Gmail intros	Google OAuth app	Pending app verification
S20 Transactional email	SendGrid key	Pending
S15 WhatsApp notify	MSG91 + Meta	Pending (~2-wk template approval)

S10-S21 — Pending

HM enrichment · intros + inbox · mock interview · tailored résumé · recruiter signup · pipeline · Nitya inbox · admin / DPDP · SEO.

S22 Deploy → Vercel (app + web) + Supabase + CI/CD

S23 Payments → deferred to v2 (manual tracking)

Keys & integrations to sort before production

In hand: APIFY_TOKEN
Needed: GOOGLE_CLIENT_ID/SECRET · SARVAM_API_KEY · SENDGRID_API_KEY · SECRET_KEY · SERVICE_SECRET
Flaky / revisit: Twilio + MSG91 OTP (verify or replace) · MSG91 WhatsApp (Meta approval) · NeverBounce (not implemented yet)

INFRASTRUCTURE & SCALE

How much load the planned setup can take

Component	Planned sizing	Comfortable MVP load*	Real bottleneck / scaling lever
Next.js app (Vercel)	Serverless + edge CDN	Auto-scales to 100s of users	Vercel plan limits · function duration
FastAPI (async)	Python 3.12 · async workers	~200–500 req/s I/O-bound	Horizontal: more instances behind LB
Supabase Postgres	Pro tier + pooler	1000s light queries/s	Connection pool size; upgrade tier
pgvector search	HNSW, cosine	<10 ms over 100k–1M vectors	Index RAM; partition by region
Aarya voice (Deepgram→Sarvam)	Pay-as-you-go	~tens of concurrent calls	STT/TTS stream concurrency quota
Agent chat (OpenRouter)	Claude via API	Rate-limit bound	Token rate limit + $ per turn
Job ingest (Apify)	Scheduled pg_cron	Batch, off-peak	Apify compute units (not user-facing)

** Planning estimates for an MVP / pilot (≈ up to 1,000 active candidates, tens of concurrent voice calls). Not yet validated under load — S22 deploy pending. The Next.js app auto-scales on Vercel (serverless); the near-term limits are external-service quotas (voice STT/TTS concurrency, OpenRouter rate + cost) and the Supabase tier (connections, Realtime sockets).*

COST & LICENCES

Tools, costs & renewals

Tool	What it's for	Plan / tier	Cost	Renews / expires
Supabase	Postgres · Auth · Storage · Realtime	Pro (annual)	$55 / yr	Yearly
Vercel	App + web hosting (edge CDN)	—	—	Not bought yet
Deepgram	Voice STT / TTS — in use now	Pay-as-you-go	—	In use
Sarvam AI	Voice STT / TTS — planned (Indian langs)	Planned	—	Not bought yet
OpenRouter	Agent LLM (Claude) — runtime	Credits	$20 top-up	On recharge
Apify	LinkedIn + job scrapers	Pay-as-you-go	$22 / 3 mo	Every 3 months
Affinda	Résumé parsing	—	—	Not bought yet
Twilio	Phone OTP / SMS	—	—	Not bought yet
MSG91	SMS / WhatsApp	—	—	Not bought yet
NeverBounce	Email verification (planned)	—	—	Not bought yet
Gmail OAuth	Recruiter intro sending	Free	Free	—
SendGrid	Transactional email	—	—	Not bought yet
Domain (hireloop.in)	Registrar / DNS	—	—	Not bought yet
Claude (Anthropic)	Dev / build assistant	Pro	$20 / mo	Monthly
Cursor	AI code editor — dev tool	Pro	$55 / yr	Yearly

Runtime spend is usage-based (OpenRouter, Deepgram → Sarvam, Apify, Twilio, MSG91). Claude + Cursor are build-time dev tools. Rows marked "Not bought yet" aren't purchased.

BEFORE PRODUCTION

Risks & must-fix items

Rotate exposed credentials

Live API credentials are currently committed inside a planning doc (PHASE_TRACKER.md). Rotate them and move every secret into Vercel / Supabase env (.env) before any deploy.

Limited end-to-end testing

The built steps work in dev, but the full journeys aren't validated e2e with live keys. Stand up staging and walk both journeys before launch.

Deployment not started (S22)

Vercel (app + web), Supabase project hardening, and GitHub Actions CI/CD still to be provisioned and wired to the custom domains.

Flaky integrations + approvals

Twilio + MSG91 OTP are unreliable — verify or pick an alternative. MSG91 WhatsApp needs Meta approval (~2 wks); Google OAuth needs verification. Start early.

NEXT

Roadmap & tracking in Linear

1

Procure keys

Google OAuth, SendGrid, Sarvam (Apify ✓)

2

E2E test staging

Walk candidate + recruiter journeys

3

Deploy (S22)

Vercel + Supabase + CI/CD

4

Pilot launch

Onboard first candidates & recruiters

Tracking model for Linear

Epics → journeys. Three epics — Candidate journey, Recruiter journey, and Platform & Infra.
Issues → steps. Each S-step (S01–S23) becomes one issue, labelled done / blocked-on-key / not-started / deferred.
Sub-tasks → the per-step checklist (test with real keys, DB check) already written in PHASE_TRACKER.md.
Labels → area:candidate · area:recruiter · area:infra · blocked:key for filtered views and burn-down.

HIRELOOP

Two AI agents. One candidate graph.

8 of 23 steps built · architecture set · intro loop + keys + deploy next.